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Process Description and Control

2

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Quick Review

- **Operating System Objectives** 目标
  - Convenience
  - Efficiency
  - Ability to evolve 能发展性
- **Evolution of Operating System**
  - Batch, multiprogramming, time-sharing, real-time, distributed…

批处理 多道 分时 实时 分布式

- **Operating System Structures**
  - Services, System call, kernel…

服务 系统调用 内核级

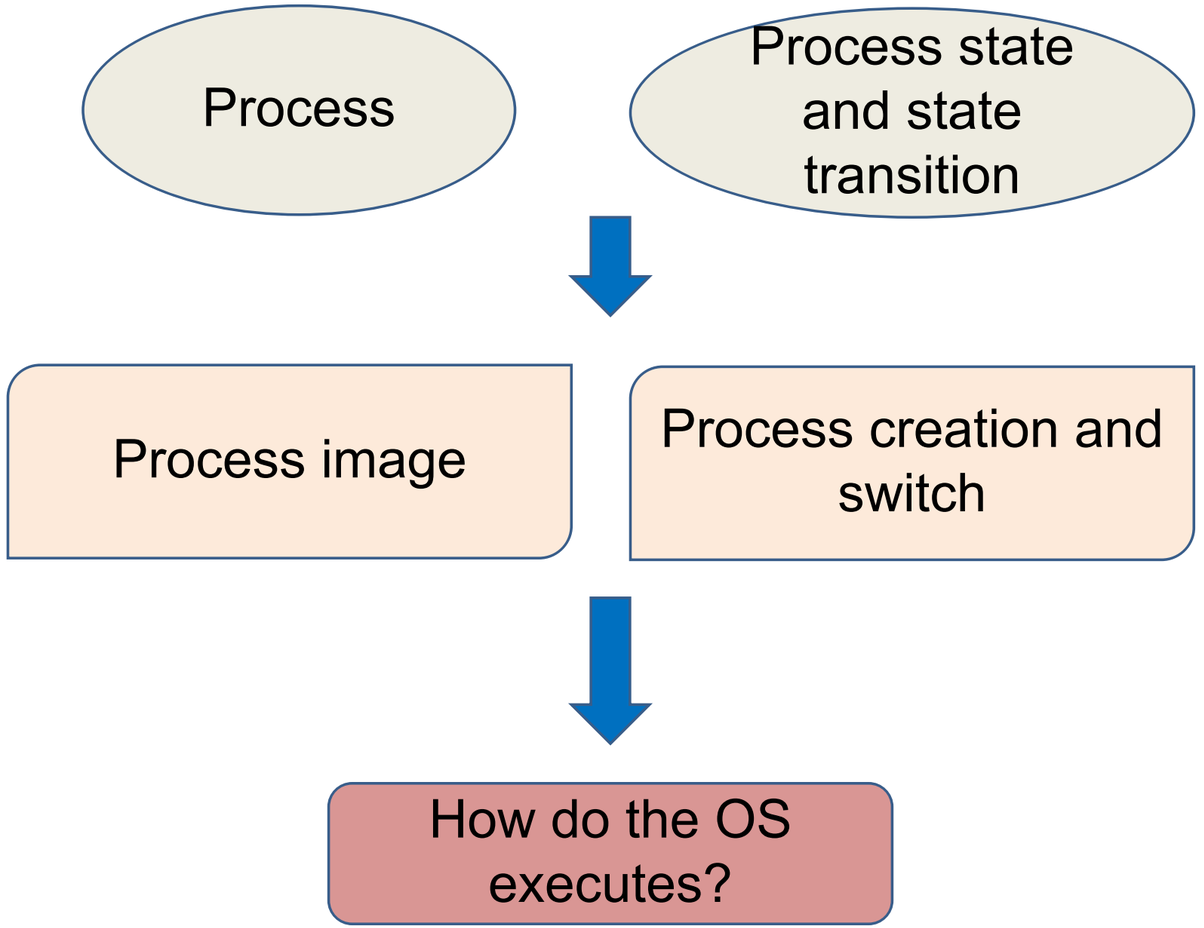
Process Management

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- Process 进程
- Threads 线程
- Scheduling 调度
- Concurrency 并发
- Synchronization and Deadlock 同步和死锁

Process Description and Control

- Process Concept
- Process State
- Process Description
- Process control
- OS Execution



OS Management of Application Execution

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- Resources are made available to multiple applications
- The processor is switched among multiple applications so all will appear to be progressing
- The processor and I/O devices can be used efficiently

2.1 Process Definitions

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Definitions

- A **program** in execution.
- An **instance** of a program running on a computer.
- The **entity** that can be assigned to and executed on a processor.
- **A unit of activity** characterized by a single sequential thread of execution, a current state, and an associated set of system resources.

实例.

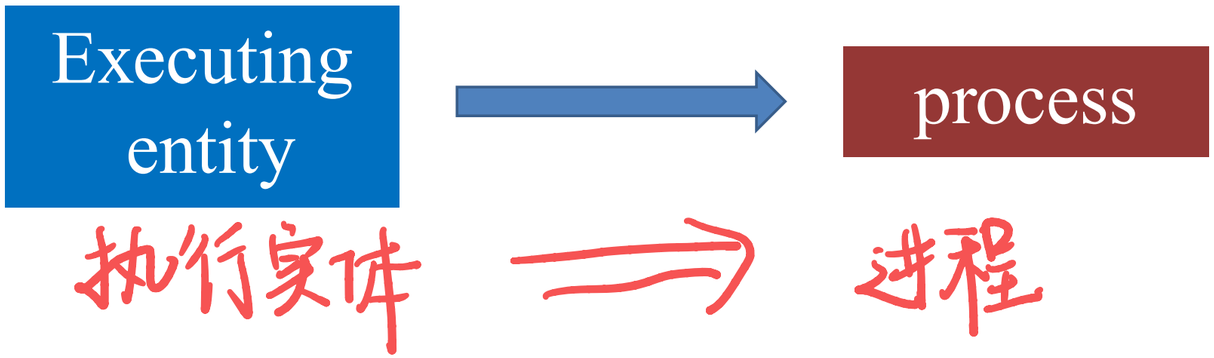
实体

以单一顺序执行流 当前状态

一组关联的系统资源

Process Elements

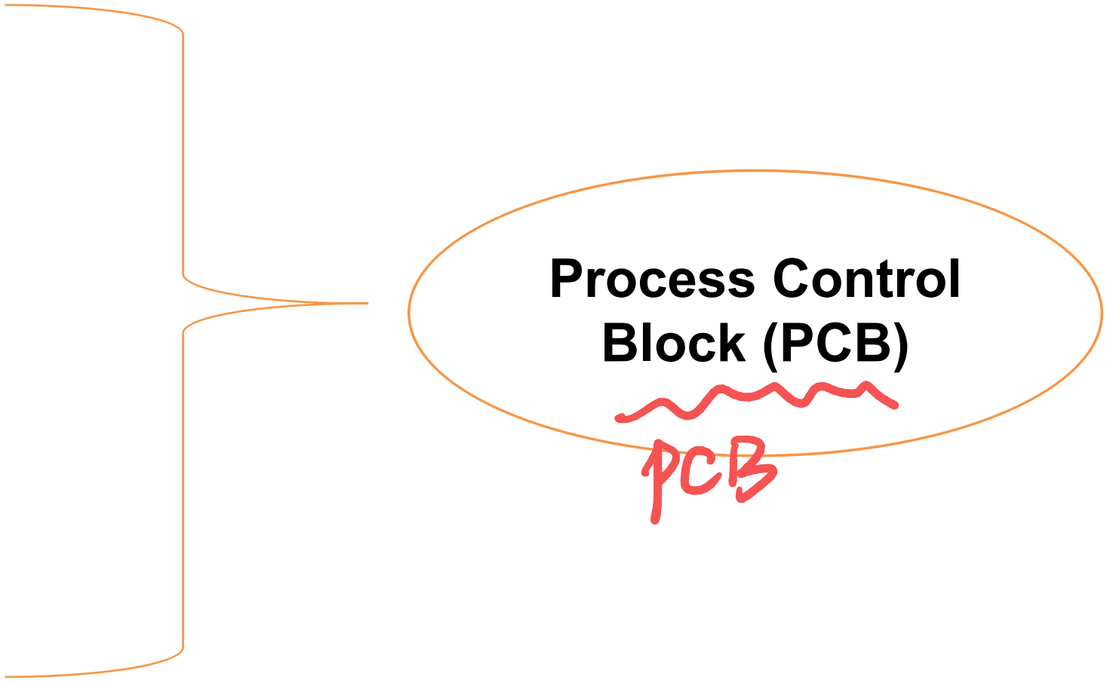
- **Program code**
  - which may be shared with other processes that are executing the same program
- **A set of data associated with that code**
  - When the **processor** begins to execute the program code, we refer to this executing entity as a process



Process Elements

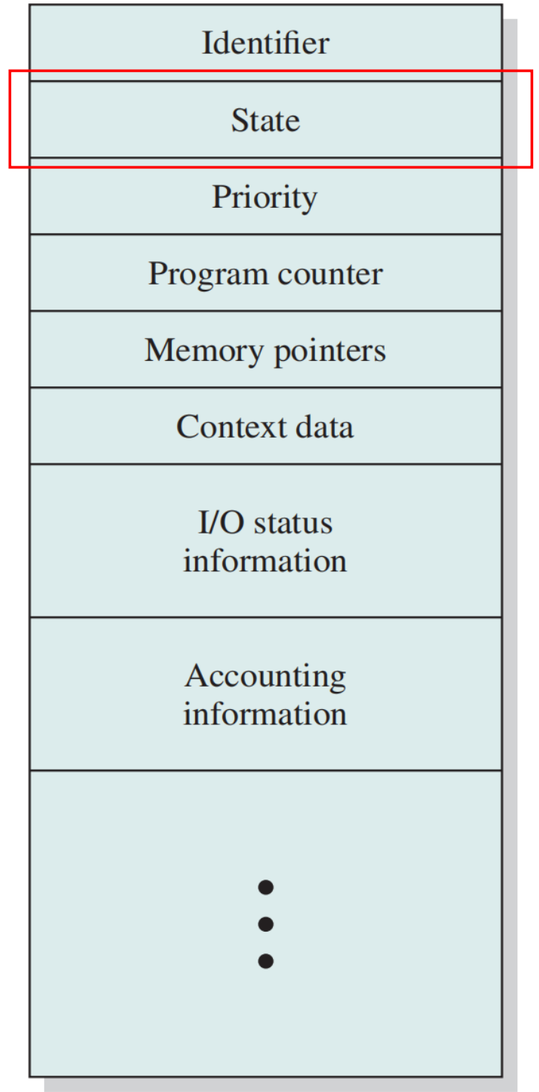
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- **While the program is executing, this process can be uniquely characterized by a number of elements, including:**
  - **identifier**
  - state
  - priority
  - program counter
  - memory pointers
  - context data
  - I/O status information
  - accounting information
  - …



Process Control Block

- Contains the process elements
- Created and managed by the operating system
- Functions:
  - It is possible to interrupt a running process and later resume execution as if the interruption had not occurred
  - Key tool that allows support for multiple processes



2.1 Process States

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Process Trace

the behavior of an individual process by listing the sequence of instructions that execute for that process

单个进程执行时的指令

the behavior of the processor can be characterized by showing how the traces of the various processes are interleaved

[时间片轮转]

交错

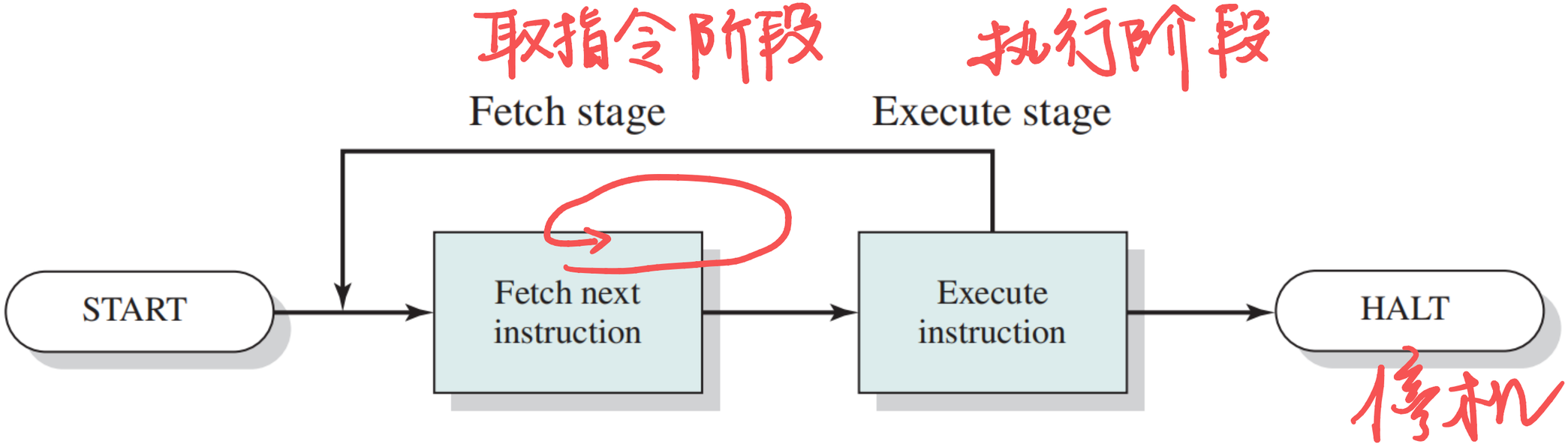
Dispatcher

进程调度/分派

small program that switches the processor from one process to another

小程序决定该让谁上CPU

Process Execution



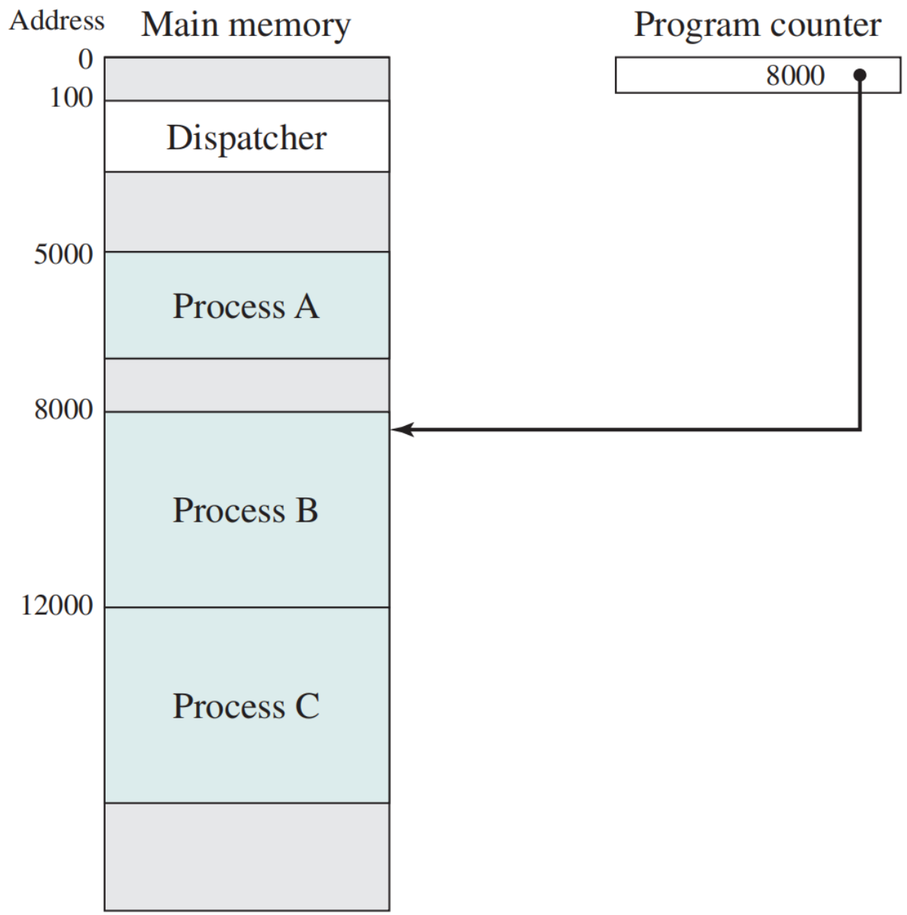
Basic Instruction Cycle

Traces of Processes

5000	8000	12000
5001	8001	12001
5002	8002	12002
5003	8003	12003
5004		12004
5005		12005
5006		12006
5007		12007
5008		12008
5009		12009
5010		12010
5011		12011

(a) Trace of process A (b) Trace of process B (c) Trace of process C

5000 = Starting address of program of process A
8000 = Starting address of program of process B
12000 = Starting address of program of process C



Combined Trace of Processes

时间片轮转

- 100 = Starting address of dispatcher program
- Shaded areas indicate execution of dispatcher process;
- Maximum six instruction cycles;

→ 100表示调度器起始

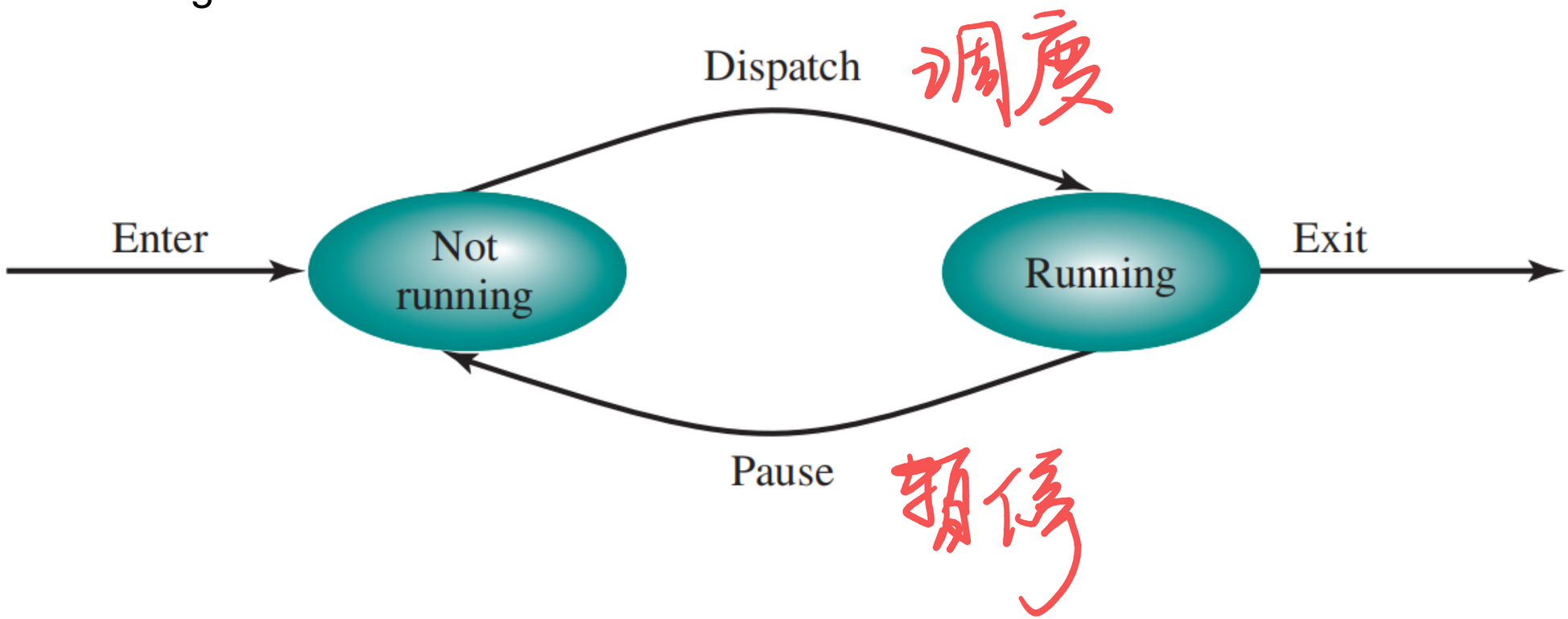
每个进程最多执行6个指令周期

1	5000	
2	5001	
3	5002	
4	5003	
5	5004	
6	5005	
----------	Time-out	
7	100	
8	101	
9	102	
10	103	
11	104	
12	105	
13	8000	
14	8001	
15	8002	
16	8003	
----------	I/O request	
17	100	
18	101	
19	102	
20	103	
21	104	
22	105	
23	12000	
24	12001	
25	12002	
26	12003	
27	12004	
28	12005	
----------	Time-out	
29	100	
30	101	
31	102	
32	103	
33	104	
34	105	
35	5006	
36	5007	
37	5008	
38	5009	
39	5010	
40	5011	
----------	Time-out	
41	100	
42	101	
43	102	
44	103	
45	104	
46	105	
47	12006	
48	12007	
49	12008	
50	12009	
51	12010	
52	12011	
----------	Time-out	

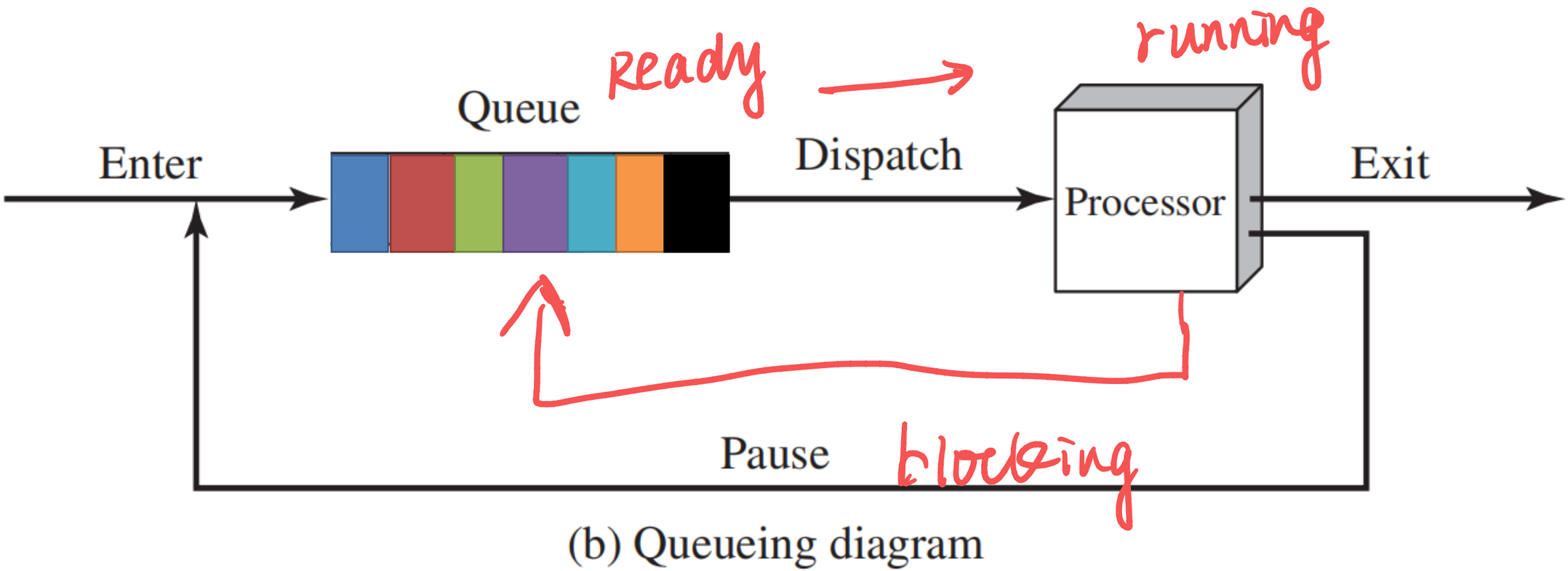
Two-State Process Model

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- **A process may be in one of two states:** 2
  - running
  - not-running



Queuing Diagram



(b) Queueing diagram

Reasons for Process Creation

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- **New batch job**
  - The OS is provided with a batch job control stream, usually on tape or disk. When the OS is prepared to take on new work, it will read the next sequence of job control commands.
- **Interactive log-on**
  - A user at a terminal logs on to the system.
- **Created by OS to provide a service**
  - The OS can create a process to perform a function on behalf of a user program, without the user having to wait (e.g., a process to control printing).
- **Spawned by existing process**
  - For purposes of modularity or to exploit parallelism, a user program can dictate the creation of a number of processes.

Process Creation

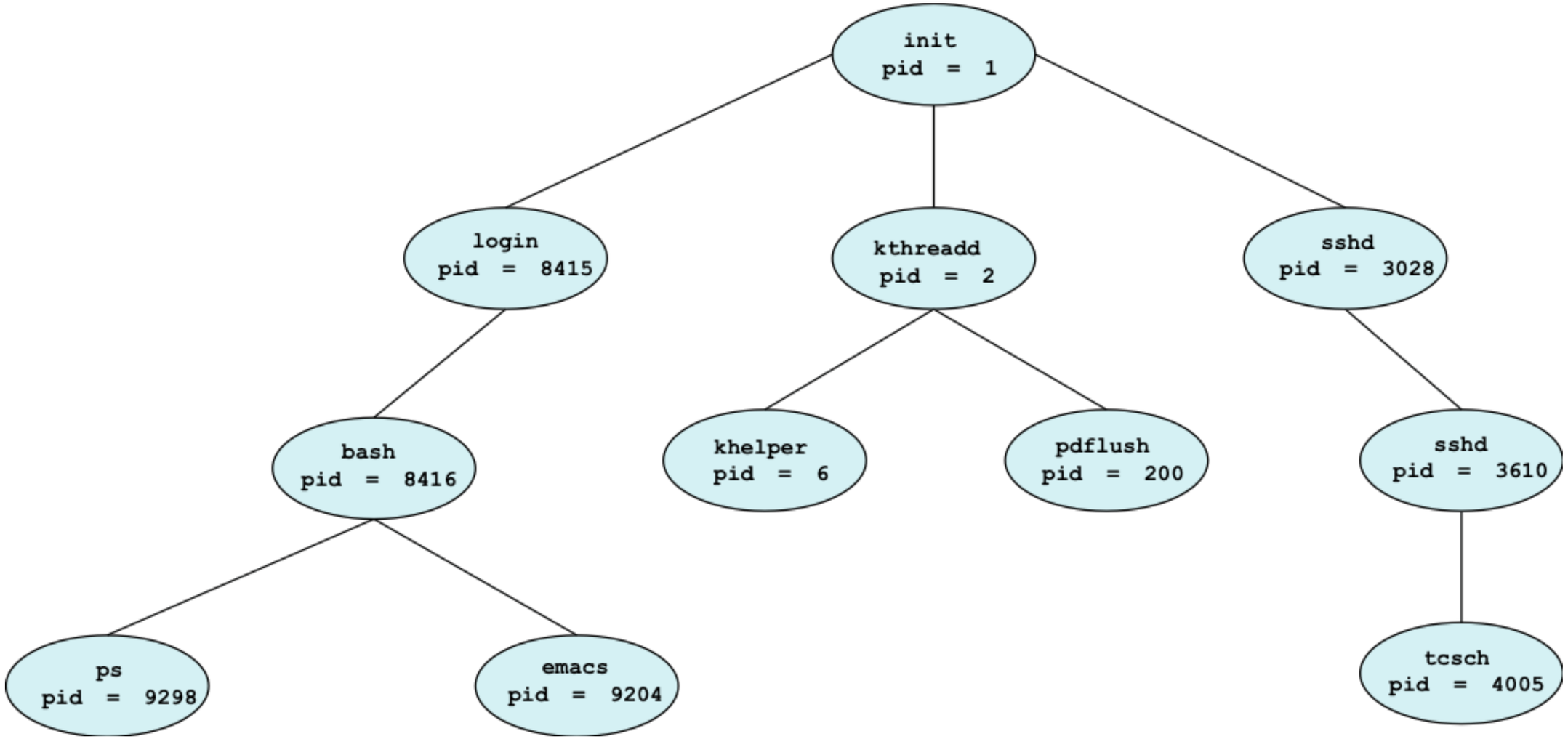
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- **Process spawning （派生）**
  - when the OS creates a process at the explicit request of another process
- **Parent process**
  - is the original, creating, process
- **Child process**
  - is the new process

显式请求

Tree of Processes in Linux

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Process Termination

- There must be a means for a process to indicate its completion
- A batch job should include a HALT instruction or an explicit OS service call for termination
- For an interactive application, the action of the user will indicate when the process is completed (e.g. log off, quitting an application)

Reasons for Process Termination

Normal completion	The process executes an OS service call to indicate that it has completed running.
Time limit exceeded	The process has run longer than the specified total time limit. There are a number of possibilities for the type of time that is measured. These include total elapsed time ("wall clock time"), amount of time spent executing, and, in the case of an interactive process, the amount of time since the user last provided any input.
Memory unavailable	The process requires more memory than the system can provide.
Bounds violation	The process tries to access a memory location that it is not allowed to access.
Protection error	The process attempts to use a resource such as a file that it is not allowed to use, or it tries to use it in an improper fashion, such as writing to a read-only file.
Arithmetic error	The process tries a prohibited computation (such as division by zero) or tries to store numbers larger than the hardware can accommodate.
Time overrun	The process has waited longer than a specified maximum for a certain event to occur.
I/O failure	An error occurs during input or output, such as inability to find a file, failure to read or write after a specified maximum number of tries (when, for example, a defective area is encountered on a tape), or invalid operation (such as reading from the line printer).
Invalid instruction	The process attempts to execute a nonexistent instruction (often a result of branching into a data area and attempting to execute the data).
Privileged instruction	The process attempts to use an instruction reserved for the operating system.
Data misuse	A piece of data is of the wrong type or is not initialized.
Operator or OS intervention	For some reason, the operator or the operating system has terminated the process (e.g., if a deadlock exists).
Parent termination	When a parent terminates, the operating system may automatically terminate all of the offspring of that parent.
Parent request	A parent process typically has the authority to terminate any of its offspring.

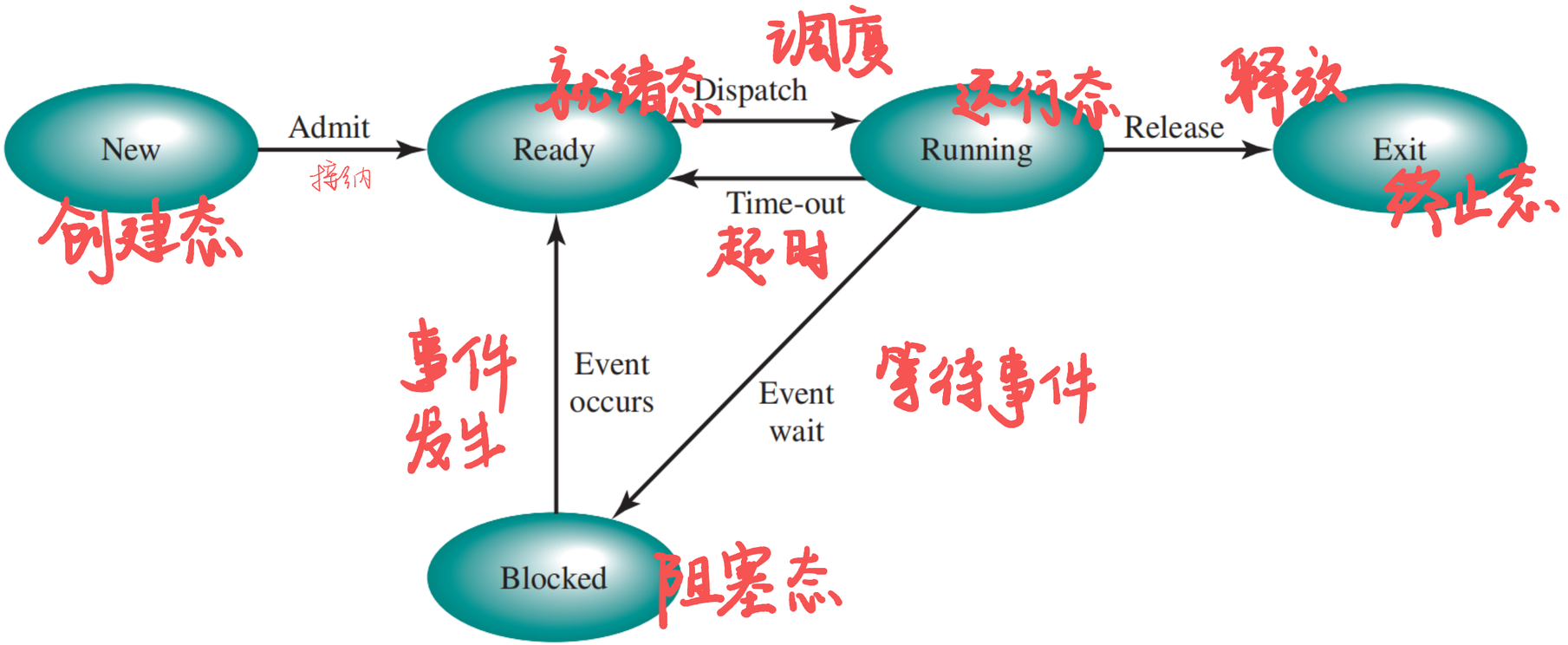
Combined Trace of Processes

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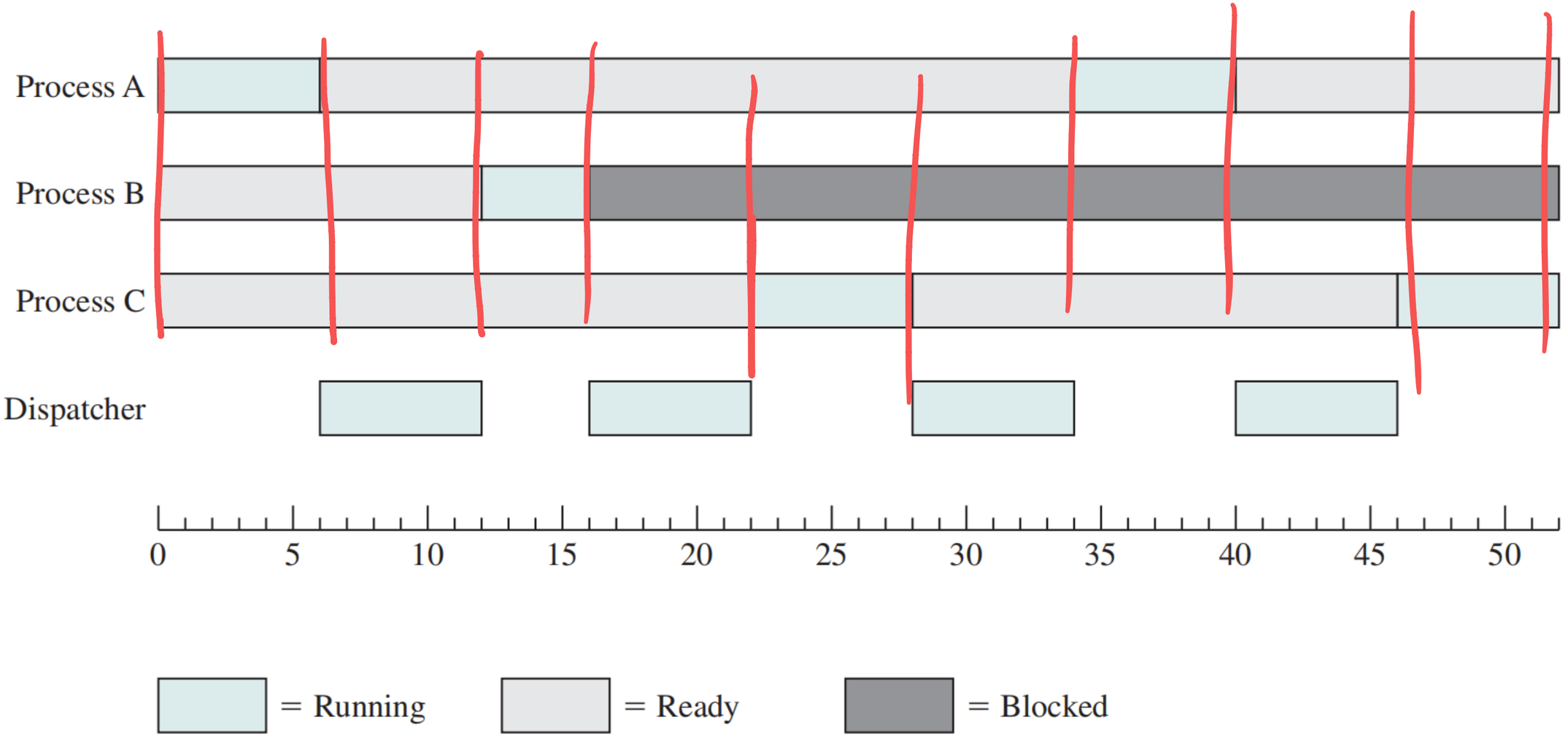
- How many states ?

1	5000
2	5001
3	5002
4	5003
5	5004
6	5005
----------Time-out	
7	100
8	101
9	102
10	103
11	104
12	105
13	8000
14	8001
15	8002
16	8003
----------I/O request	
17	100
18	101
19	102
20	103
21	104
22	105
23	12000
24	12001
25	12002
26	12003
27	12004
28	12005
----------Time-out	
29	100
30	101
31	102
32	103
33	104
34	105
35	5006
36	5007
37	5008
38	5009
39	5010
40	5011
----------Time-out	
41	100
42	101
43	102
44	103
45	104
46	105
47	12006
48	12007
49	12008
50	12009
51	12010
52	12011
----------Time-out	

Five-State Process Model

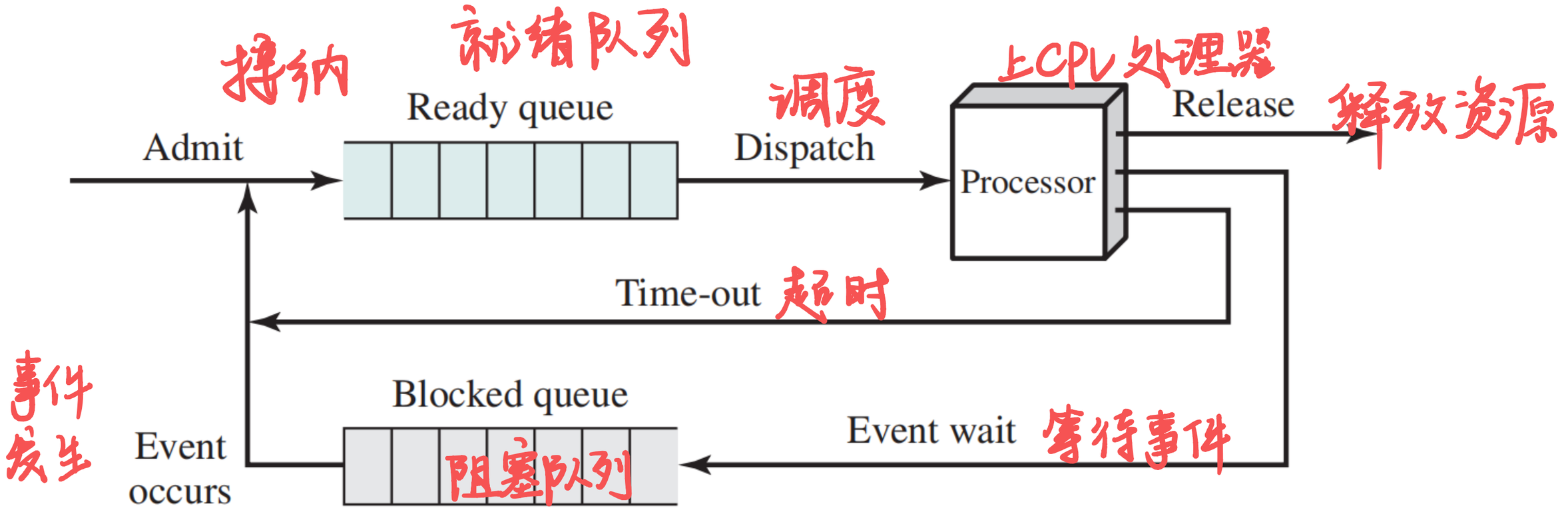


Process States for the Trace



Transition of each process among the states.

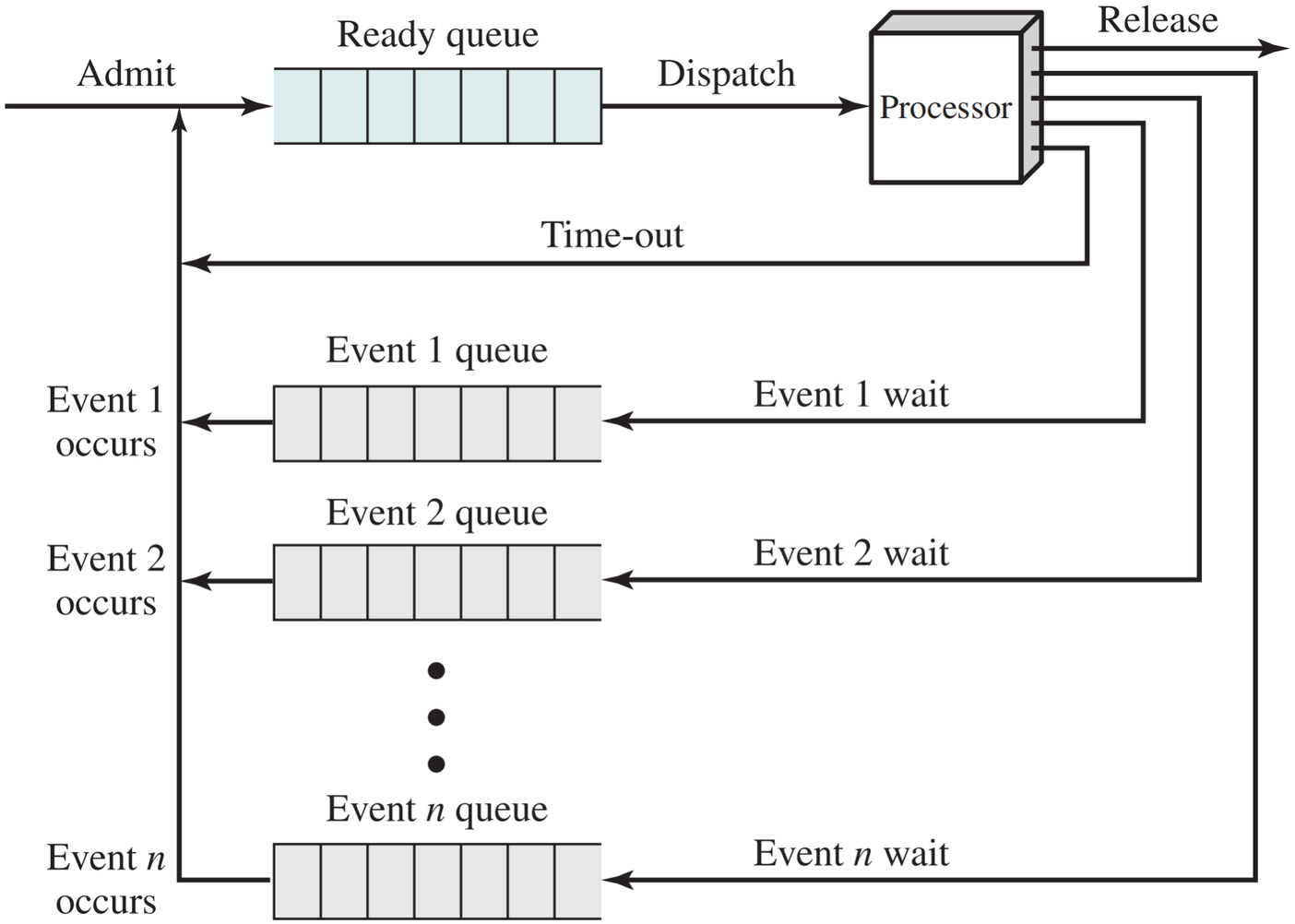
Using Two Queues（双队列模型）



(a) Single blocked queue

Multiple Blocked Queues

多阻塞队列



(b) Multiple blocked queues

The processor is so much faster than I/O that it will be common for **all of the processes in memory to be waiting for I/O**.

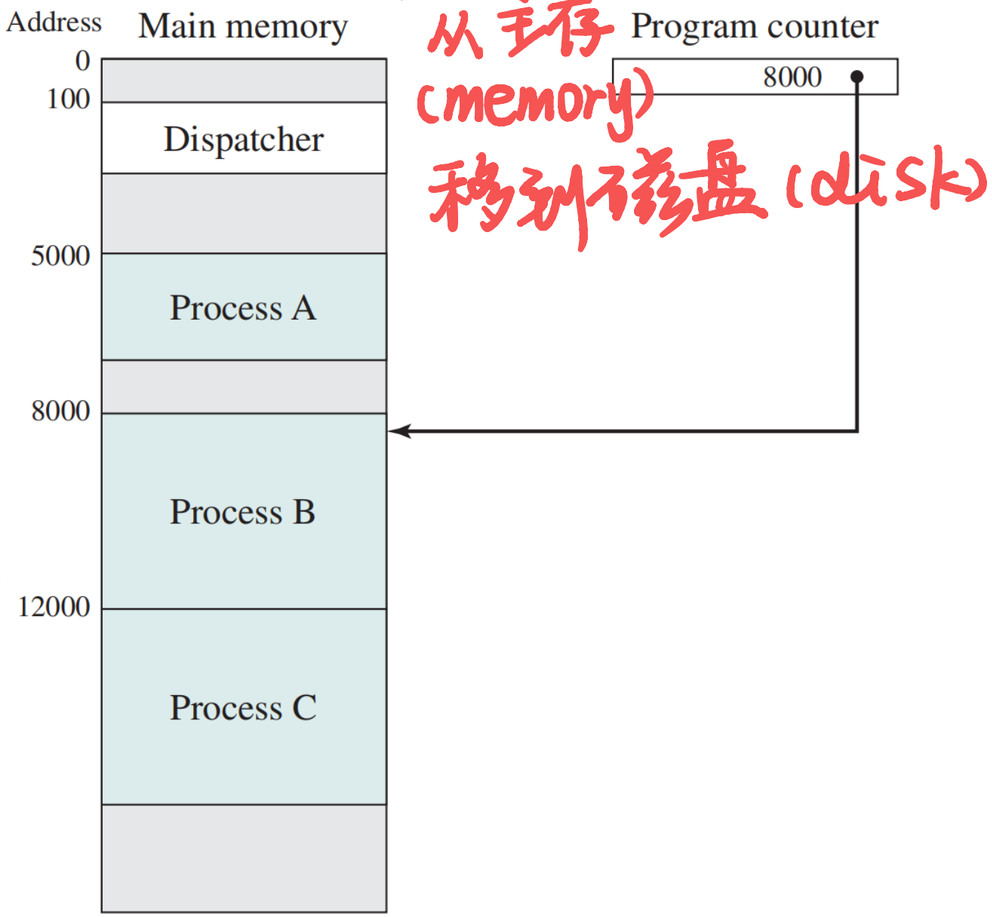
What to do ?

Suspended Processes 挂起进程

- Swapping (交换)
- involves moving part of all of a process from main memory to disk
- when none of the processes in main memory is in the Ready state, the OS swaps one of the blocked processes out on to disk into a suspend queue

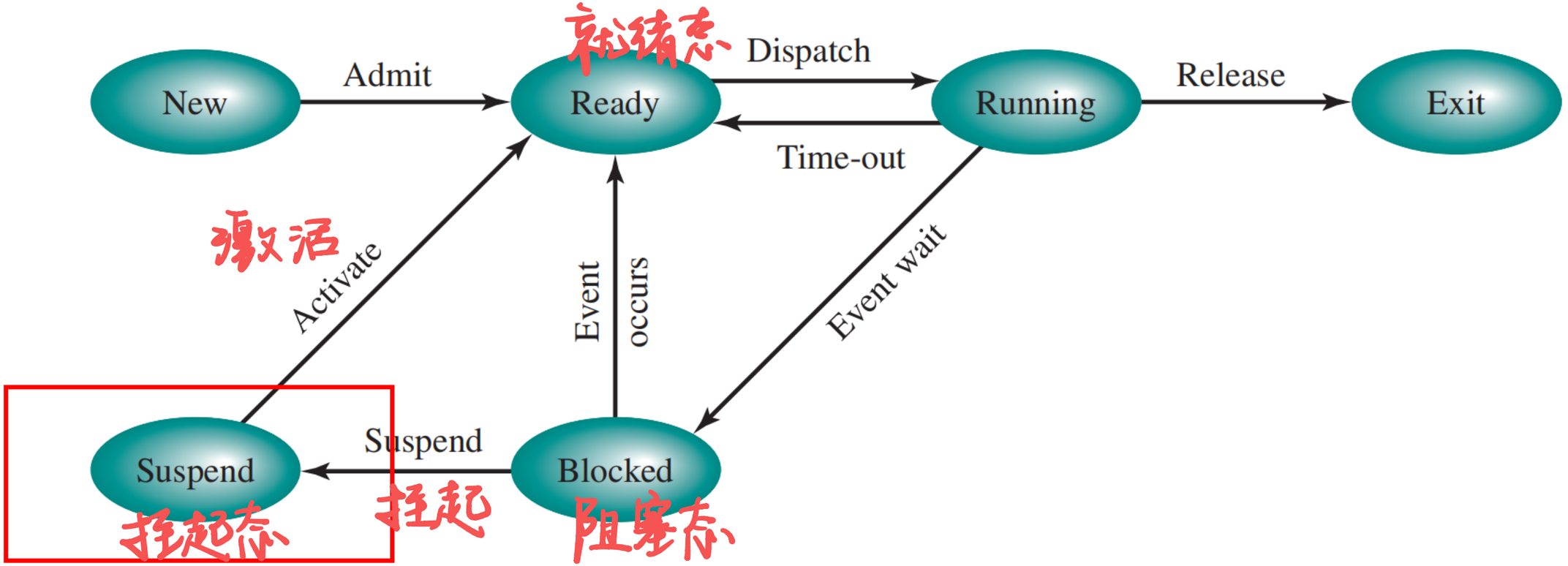
将进程的部分或全部数据从主存(memory)移到磁盘(disk)

将某个阻塞进程交换到磁盘的挂起队列中

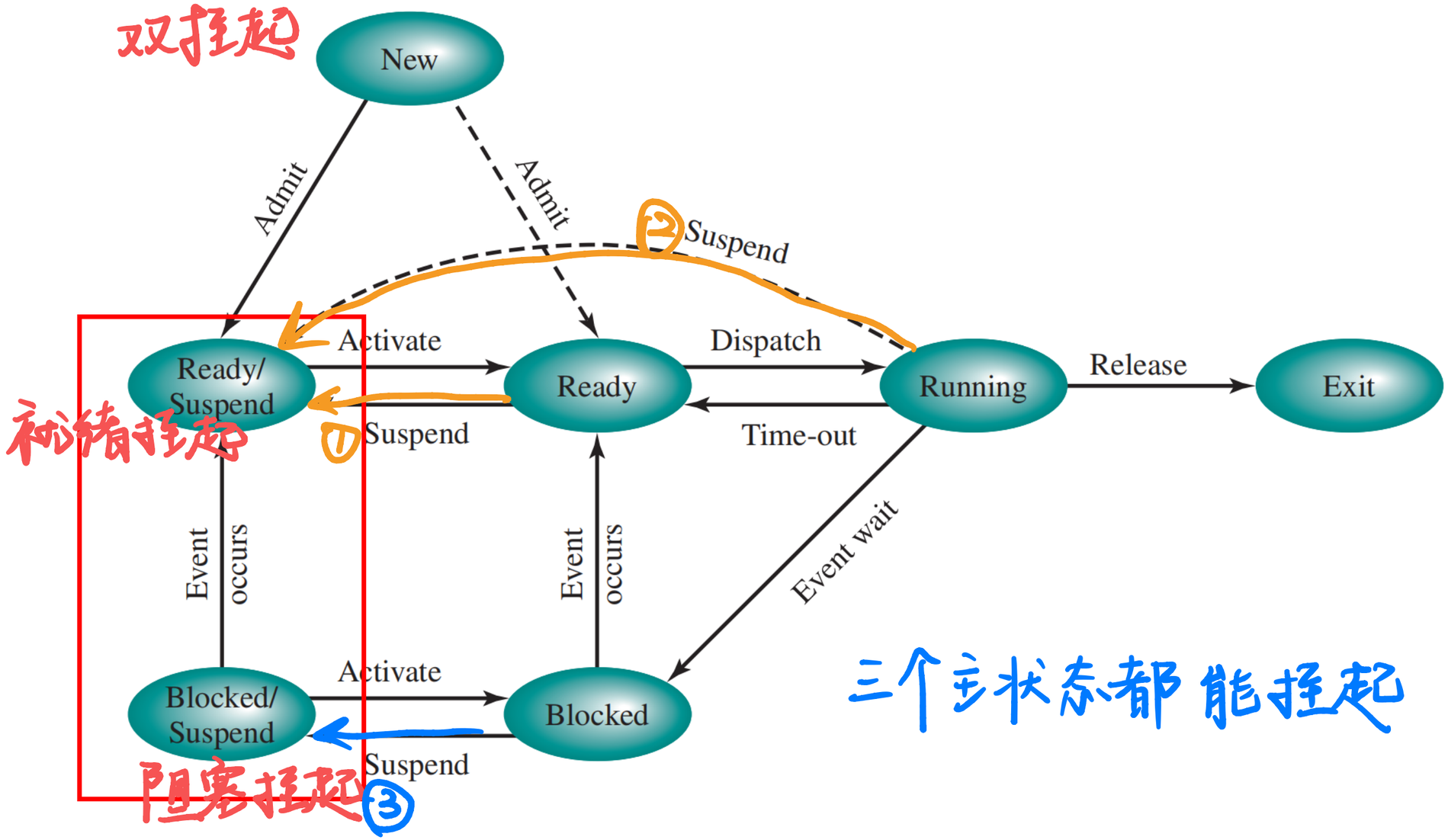


One Suspend State (单挂起)

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Two Suspend States



Reasons for Process Suspension

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Swapping 交换	The OS needs to release sufficient main memory to bring in a process that is ready to execute. 释放足够的主存以调入就绪态的进程
Other OS reason	The OS may suspend a background or utility process or a process that is suspected of causing a problem. 后台 实用进程
Interactive user request 交互式用户请求	A user may wish to suspend execution of a program for purposes of debugging or in connection with the use of a resource. 调试
Timing 计时	A process may be executed periodically (e.g., an accounting or system monitoring process) and may be suspended while waiting for the next time interval.
Parent process request 父进程	A parent process may wish to suspend execution of a descendent 子进程 to examine or modify the suspended process, or to coordinate 协调 the activity of various descendants.

Characteristics of a Suspended Process

特征

- The process is not immediately available for execution

①无法立即执行

- The process was placed in a suspended state by an agent: either itself, a parent process, or the OS, for the purpose of preventing its execution

智能体

②由自身，父进程或操作系统挂起，为了阻止其运行

- The process may or may not be waiting on an event

③可能(也可能不是)在等待事件

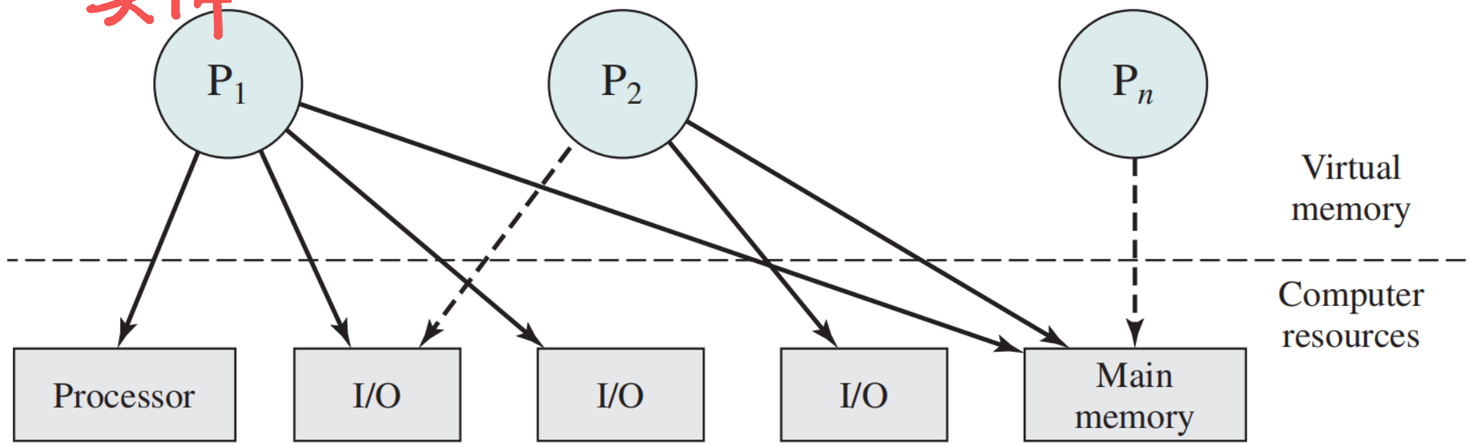
- The process may not be removed from this state until the agent explicitly orders the removal

明确

(除非主体明确下达移除命令，否则进程无法退出挂起态)

2.3 Process Description

- OS schedules and dispatches processes for execution by the processor, allocates resources to processes, and responds to requests by user processes for basic services.
- OS as that entity that manages the use of system resources by processes.



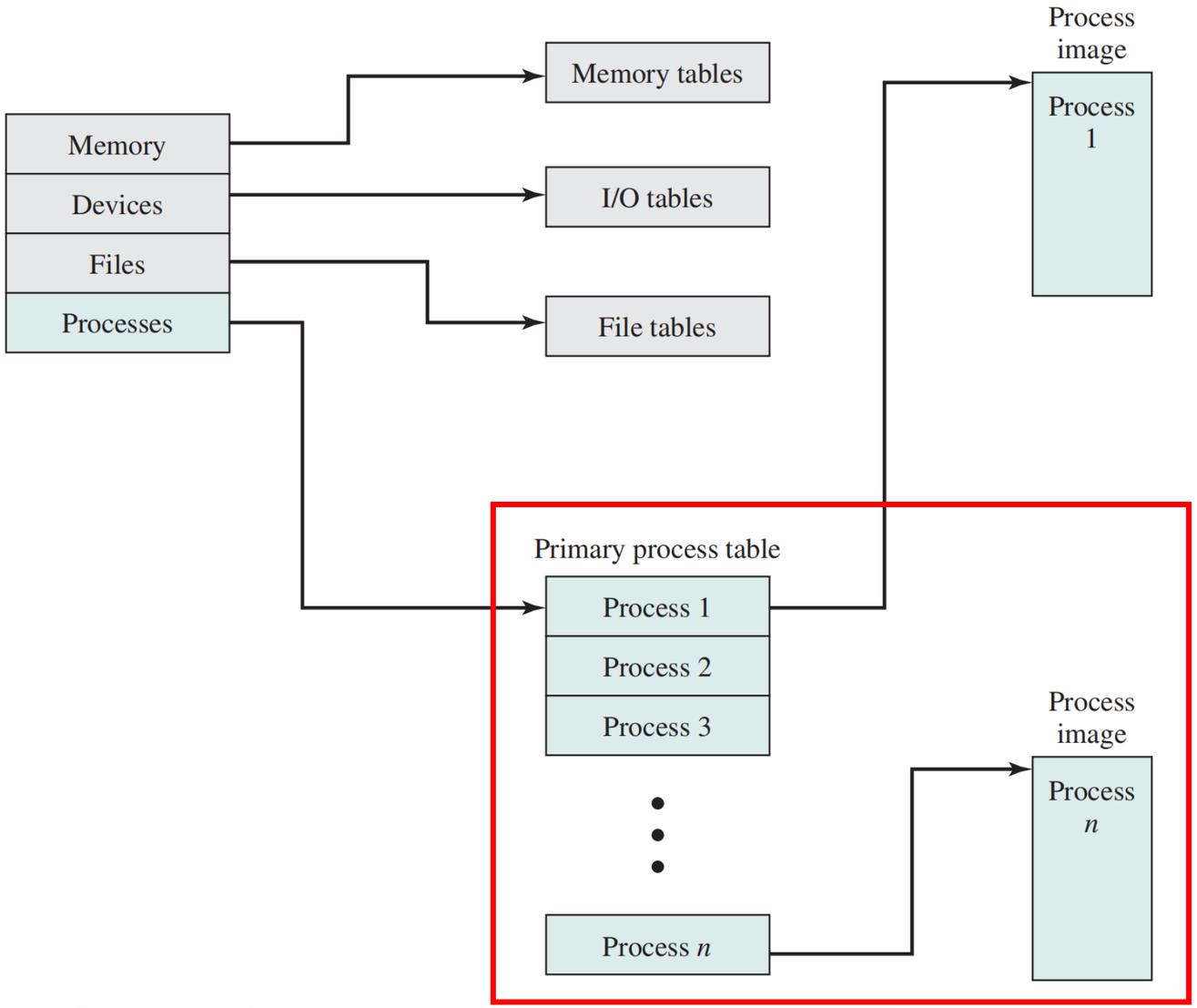
What information does the OS need to control processes and manage resources for them?

OS Control Structures

Process table

- Must be maintained to manage processes
- There must be some reference to memory, I/O, and files, directly or indirectly
- The tables themselves must be accessible by the OS and therefore are subject to memory management

可被访问的



General Structure of Operating System Control Tables

Process Control Structures

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- **To manage and control a process the OS must know:**
  - where the process is located
  - the attributes (PID, State) of the process that are necessary for its management

属性

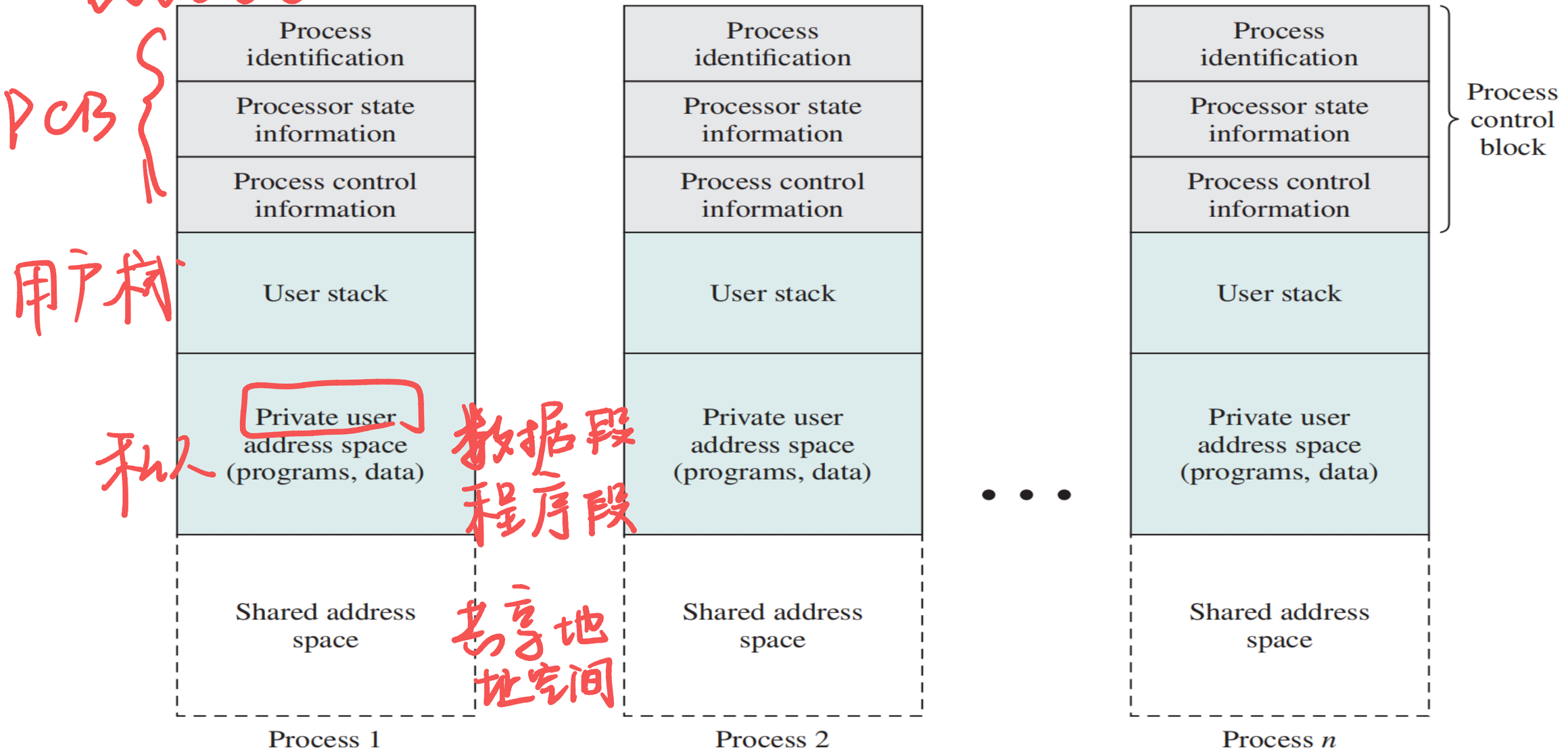
PCB

Process Image

进程映像

- **What is the physical manifestation (物理表示) of a process?**
  - A process must include a program or set of programs to be executed
  - A process will consist of at least sufficient memory to hold the programs and data of that process
  - The execution of a program typically involves a stack that is used to keep track of procedure calls and parameter passing between procedures
  - Each process has associated with it a number of attributes that are used by the OS for process control
- **The collection of program, data, stack, and attributes is referred to as the process image**
- **Process image location will depend on the memory management scheme being used**

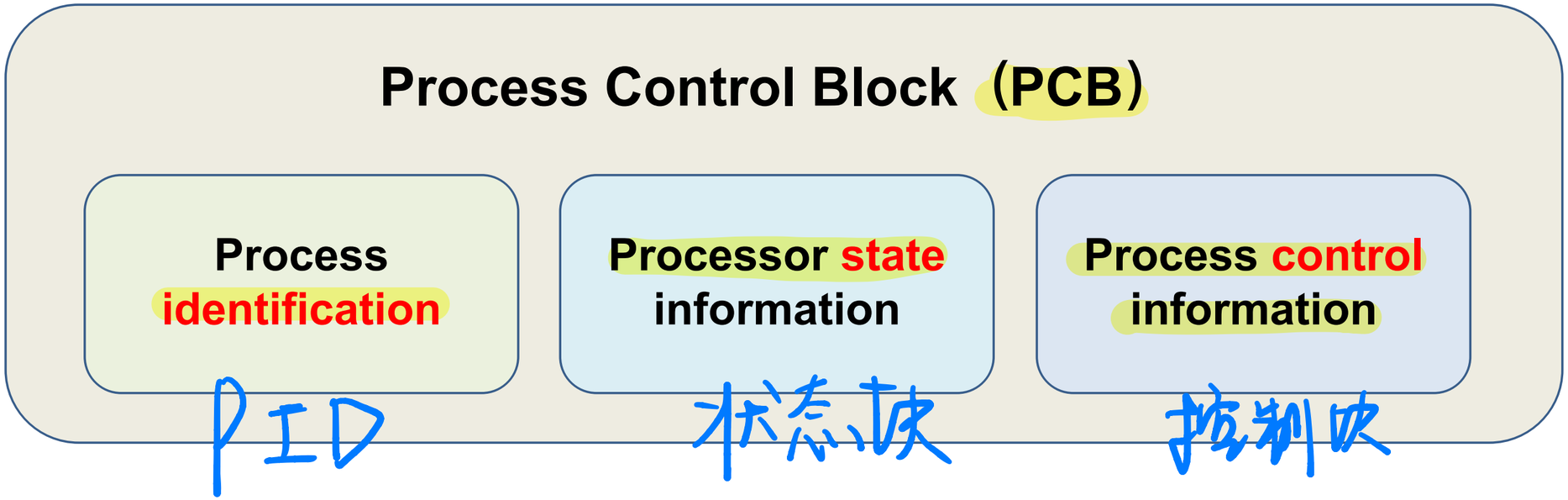
Typical Elements of a Process Image



Process Control Block (PCB)

数据结构

- **The most important data structure in an OS**
  - contains all of the information about a process that is needed by the OS
  - blocks are read and/or modified by every module in the OS
  - defines the state of the OS



PCB-Process Identification（进程标识符）

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- **Each process is assigned a unique numeric identifier**
  - otherwise there must be a mapping that allows the OS to locate the appropriate tables based on the process identifier
- **Many of the tables controlled by the OS may use process identifiers to cross-reference process tables** 通过PID交插引用
- **Memory tables may be organized to provide a map of main memory with an indication of which process is assigned to each region**
  - similar references will appear in I/O and file tables

主存映射[有地址]

- **When processes communicate with one another, the process identifier informs the OS of the destination of a particular communication** 反流
- **When processes are allowed to create other processes, identifiers indicate the parent and descendants of each process（父亲和后代进程）**

标识

PCB-Processor State Information

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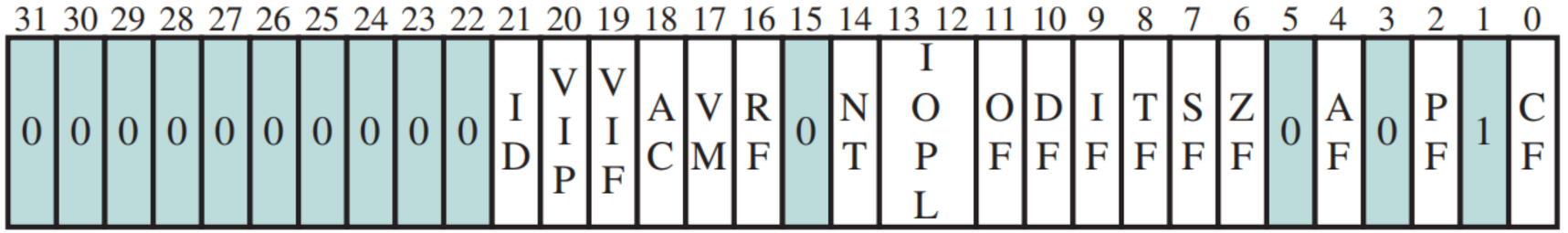
- **Consists of the contents of processor registers**
  - user-visible registers
  - stack pointers
  - control and status registers： PC，condition code， status info.

处理寄存器

- **All processor design include program status word (PSW)**
  - contains condition codes （条件码） plus other status information
  - EFLAGS register is an example of a PSW used by any OS running on an x86 processor

程序状态字

X86 EFLAGS Register



X ID = Identification flag
X VIP = Virtual interrupt pending
X VIF = Virtual interrupt flag
X AC = Alignment check
X VM = Virtual 8086 mode
X RF = Resume flag
X NT = Nested task flag
X IOPL = I/O privilege level
S OF = Overflow flag

C DF = Direction flag
X IF = Interrupt enable flag
X TF = Trap flag
S SF = Sign flag
S ZF = Zero flag
S AF = Auxiliary carry flag
S PF = Parity flag
S CF = Carry flag

S Indicates a Status Flag
C Indicates a Control Flag
X Indicates a System Flag
Shaded bits are reserved

PCB-P

- The
coor
- Take

ntrol and

Process Control Information

Scheduling and State Information

This is information that is needed by the operating system to perform its scheduling function. Typical items of information include:

- **Process state:** Defines the readiness of the process to be scheduled for execution (e.g., running, ready, waiting, halted).
- **Priority:** One or more fields may be used to describe the scheduling priority of the process. In some systems, several values are required (e.g., default, current, highest allowable).
- **Scheduling-related information:** This will depend on the scheduling algorithm used. Examples are the amount of time that the process has been waiting and the amount of time that the process executed the last time it was running.
- **Event:** Identity of event the process is awaiting before it can be resumed.

Data Structuring

A process may be linked to other process in a queue, ring, or some other structure. For example, all processes in a waiting state for a particular priority level may be linked in a queue. A process may exhibit a parent–child (creator–created) relationship with another process. The process control block may contain pointers to other processes to support these structures.

Interprocess Communication

Various flags, signals, and messages may be associated with communication between two independent processes. Some or all of this information may be maintained in the process control block.

Process Privileges

Processes are granted privileges in terms of the memory that may be accessed and the types of instructions that may be executed. In addition, privileges may apply to the use of system utilities and services.

Memory Management

This section may include pointers to segment and/or page tables that describe the virtual memory assigned to this process.

Resource Ownership and Utilization

Resources controlled by the process may be indicated, such as opened files. A history of utilization of the processor or other resources may also be included; this information may be needed by the scheduler.

Role of the Process Control Block

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The most important data structure in an OS

- contains all of the information about a process that is needed by the OS
- blocks are read and/or modified by virtually every module in the OS
- defines the state of the OS

Difficulty is not access, but protection

设计难点不在于访问，而在于保护

- a bug in a single routine could damage process control blocks, which could destroy the system's ability to manage the affected processes
- a design change in the structure or semantics of the process control block could affect a number of modules in the OS

单例程上的漏洞

设计变更

2.4 Process Control

- Modes of execution
- Process creation
- Process switching

Modes of Execution

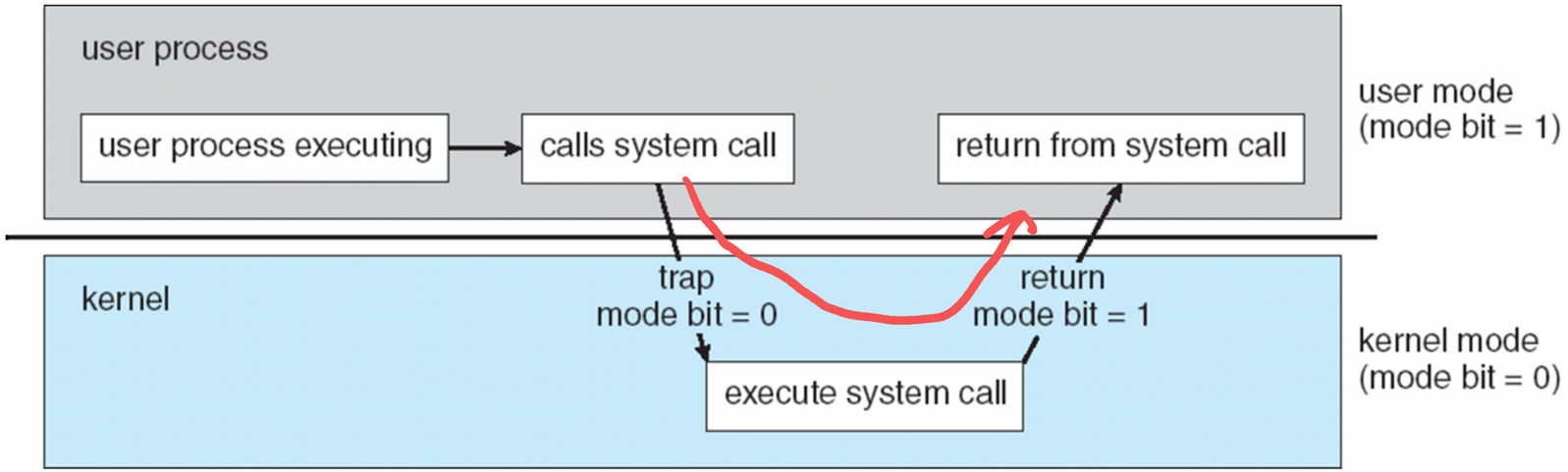
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User Mode 用户模式（用户态）

- less-privileged mode
- user programs typically execute in this mode
- PSW-current privilege level (CPL)

System Mode 系统模式（内核态）

- more-privileged mode
- also referred to as control mode or kernel mode
- kernel of the operating system



Typical Functions of an Operating System Kernel

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内核态功能

① Process Management

- Process creation and termination
- Process scheduling and dispatching
- Process switching
- Process synchronization and support for interprocess communication
- Management of process control blocks

② Memory Management

- Allocation of address space to processes
- Swapping
- Page and segment management

③ I/O Management

- Buffer management
- Allocation of I/O channels and devices to processes

④ Support Functions

- Interrupt handling
- Accounting
- Monitoring

Step of Process Creation

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- **Once the OS decides to create a new process**
  - ① assigns a unique process identifier to the new process
  - ② allocates space for the process
  - ③ initializes the PCB
  - ④ sets the appropriate linkages
  - ⑤ creates or expands other data structures

① 为新进程分配唯一的进程标识符

② 为进程分配存储空间

③ 初始化 PCB

④ 设置相应的链接

⑤ 扩展或创建其他数据结构

Process Switching

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- **A process switch may occur any time that the OS has gained control from the currently running process.**
  - Possible events giving OS control are: Interrupt, Trap and Supervisor call

中断 陷阱 管态调用

- **Interrupt**
  - Due to some external event, which is independent of the currently running process
    - clock interrupt
    - I/O interrupt
    - memory fault

外部

- **Trap**

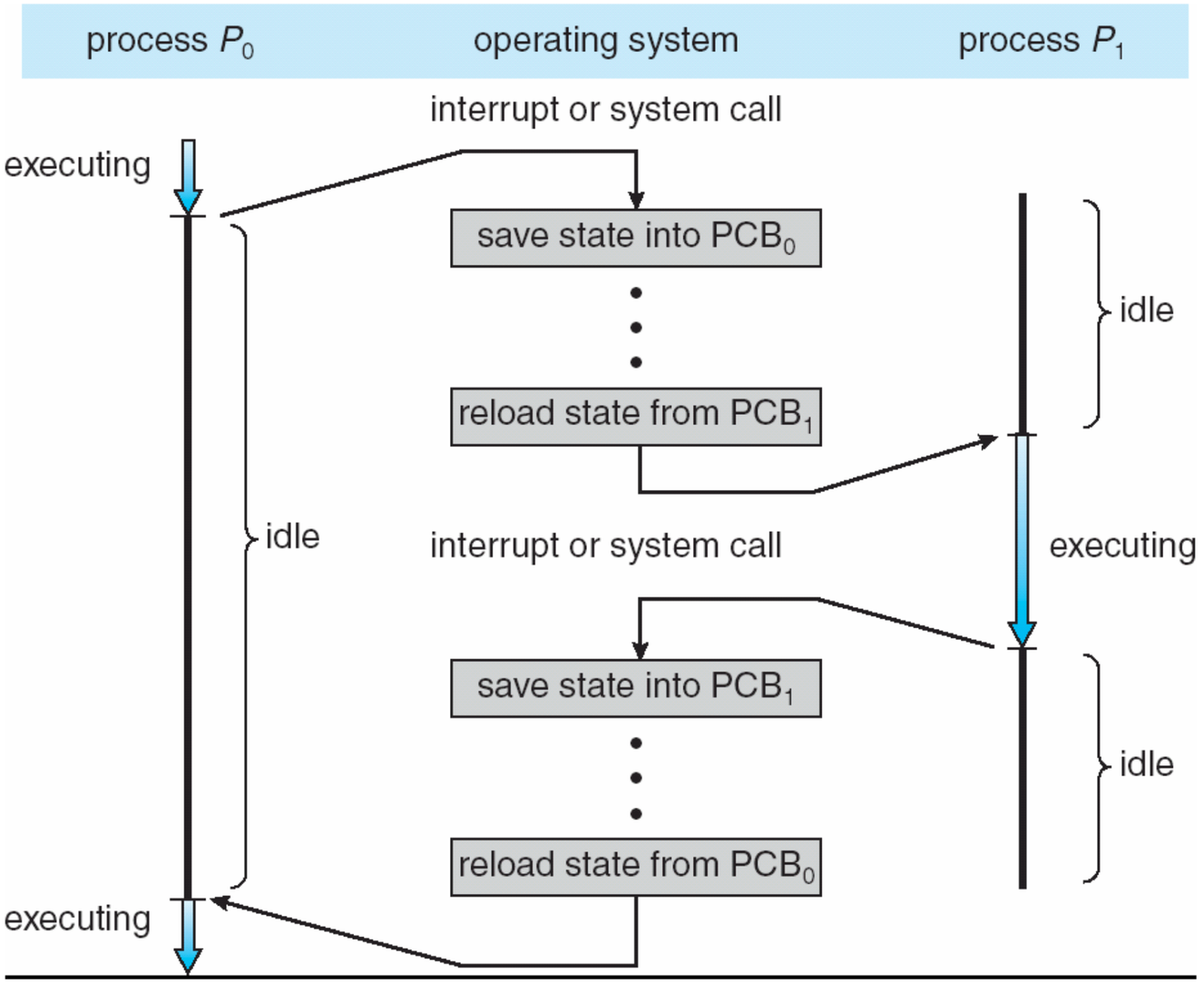
异常

  - An error or exception condition generated within the currently running process
  - OS determines if the condition is fatal

致命

    - moved to the Exit state and a process switch occurs
    - action will depend on the nature of the error

Example of Process Switch

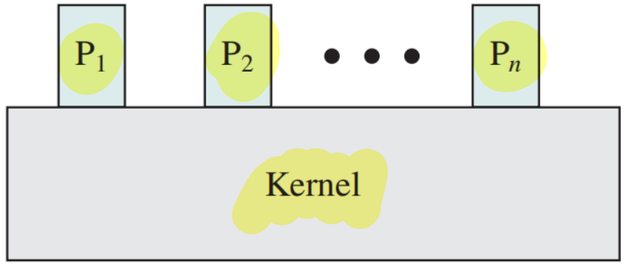


Change of Process State

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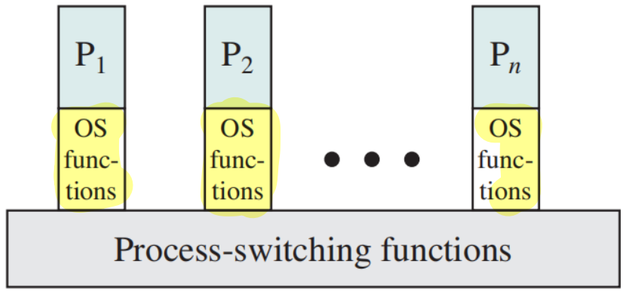
- **The steps in a full process switch are:**
  1. save the context of the processor
  2. update the process control block of the process currently in the Running state
  3. move the process control block of this process to the appropriate queue
  4. select another process for execution
  5. update the process control block of the process selected
  6. update memory management data structures
  7. restore the context of the processor to that which existed at the time the selected process was last switched out

Execution of the OS



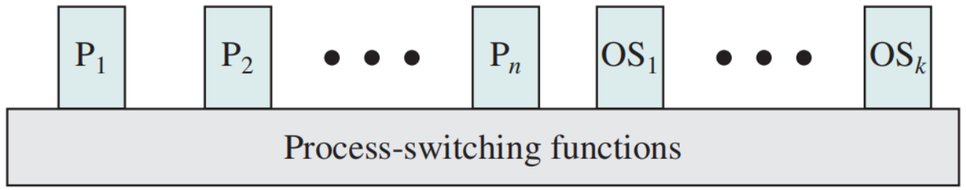
(a) Separate kernel

Nonprocess Kernel 非进程化内核



(b) OS functions execute within user processes

Execution within User Processes



(c) OS functions execute as separate processes

Process-Based Operating System

Relationship between Operating System and User Processes

Execution of the OS

Nonprocess kernel

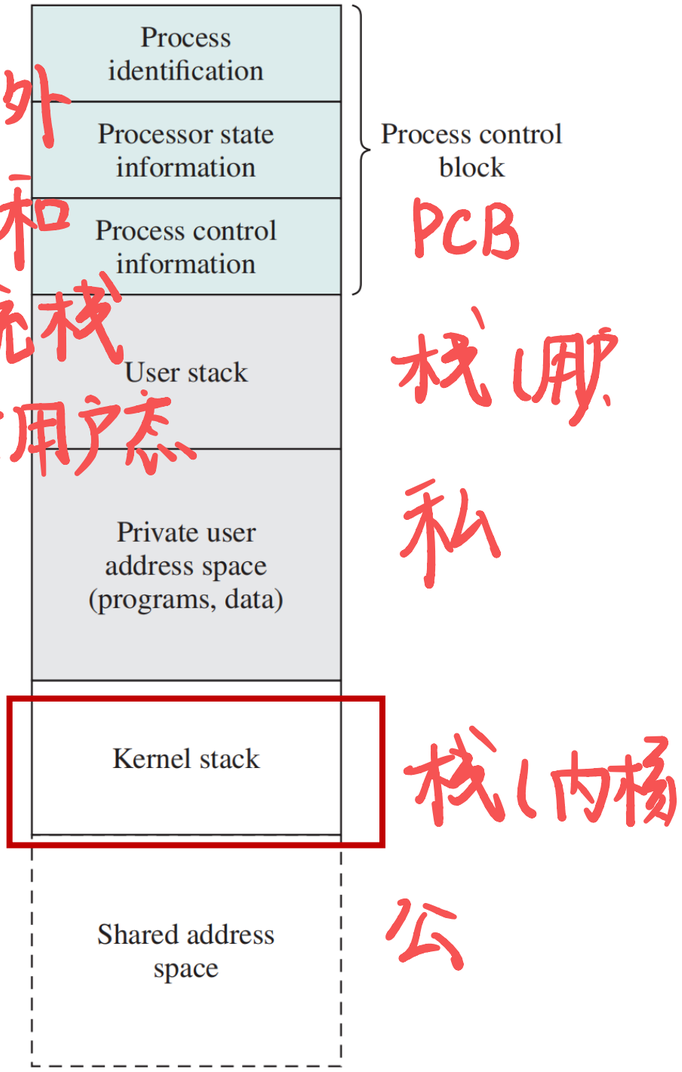
- Kernel is outside of any process
- OS has its own region of memory and system stack
- Concept of process applies only to user program.

Execution within user processes

- Small computers
- OS is viewed as a collection of routines（例程）
- When an interrupt occurs, mode switching passes control to OS, and performs mode switch rather the process switch.

Process-based OS

- Multiprocessor environment
- OS is viewed as a collection of system processes
- Clean interfaces between modules



Unix

System processes run in kernel mode

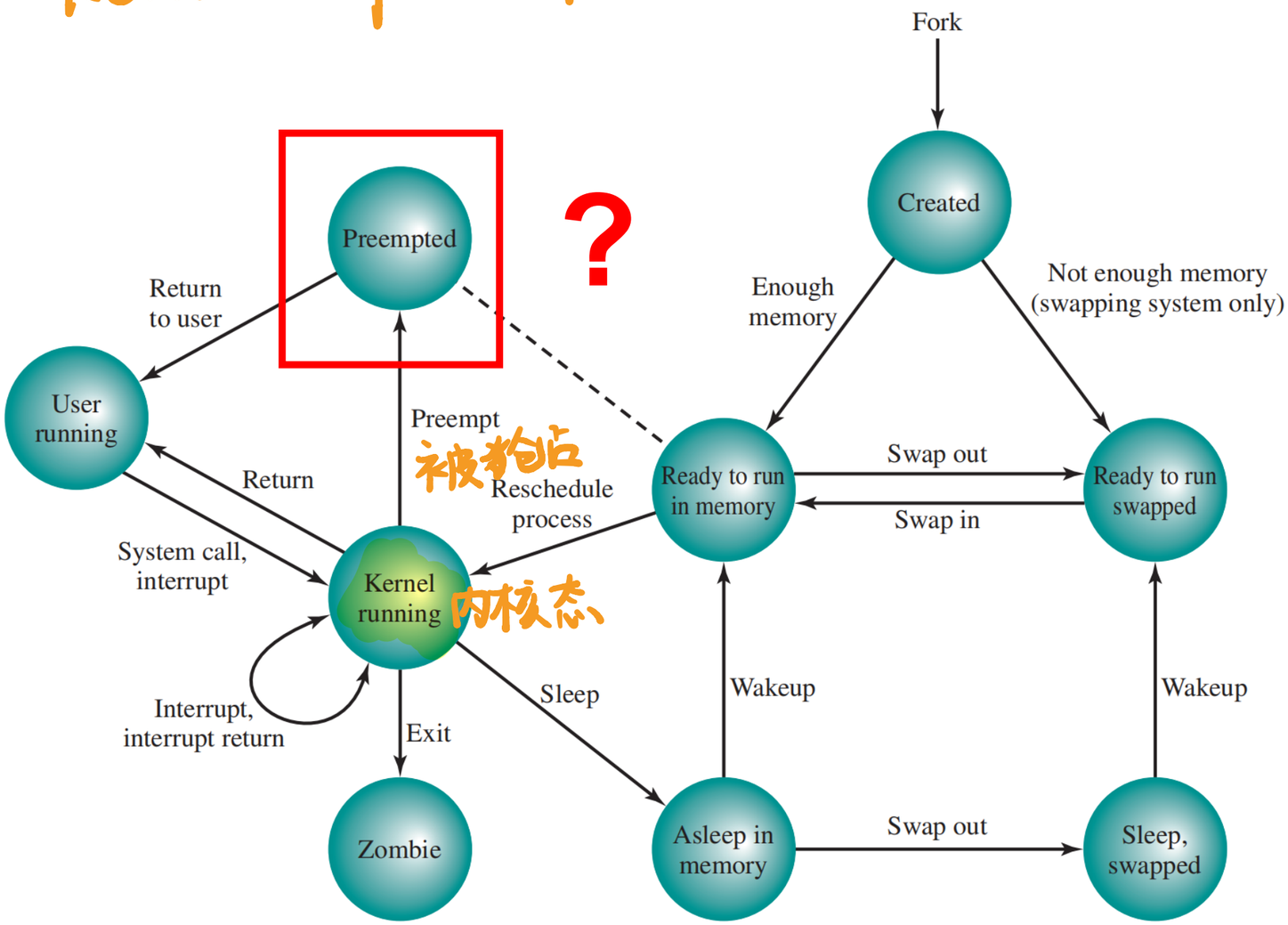
- executes operating system code to perform administrative and housekeeping functions

User Processes

- operate in user mode to execute user programs and utilities
- operate in kernel mode to execute instructions that belong to the kernel
- enter kernel mode by issuing a system call, when an exception is generated, or when an interrupt occurs

UNIX Process State

- **Preempted (被抢占)**: Process is returning from kernel to user mode, but the kernel preempts it and does a process switch to schedule another process.



Process Creation in Unix

- Process creation is by means of the kernel system call, fork( )
- This causes the OS, in Kernel Mode,

1. Allocate a slot in the process table for the new process
2. Assign a unique process ID to the child process
3. Make a copy of the process image of the parent, with the exception of any shared memory
4. Increments counters for any files owned by the parent, to reflect that an additional process now also owns those files
5. Assigns the child process to the Ready to Run state
6. Returns the ID number of the child to the parent process, and a 0 value to the child process



return 0;

After Creation

- **After creating the process the Kernel can do one of the following, as part of the dispatcher routine:**
  - stay in the parent process 继续运行父进程
  - transfer control to the child process 将控制权移给子进程
  - transfer control to another process

- **When the return from the fork occurs, the return parameter is tested. If the value is zero, then this is the child process, and a branch can be executed to the appropriate user program to continue execution. If the value is nonzero, then this is the parent process, and the main line of execution can continue.**

Summary

- **Process Definition & Elements**
- **Process states model, eg 2,5,7**
- **Process description**
  - Process image:
    - ✔ PCB
    - ✔ User stack
    - ✔ Private user address space (programs, data)
    - ✔ Shared address space
  - PCB: identification, state info and control info
- **Process execution**
  - Modes of execution: kernel and user
  - Change of process state, 7 steps
- **Process in Unix**
  - Preempted
  - Creation

